INGUINAL CANAL

Department of Anatomy

The abdomen (2)

First part

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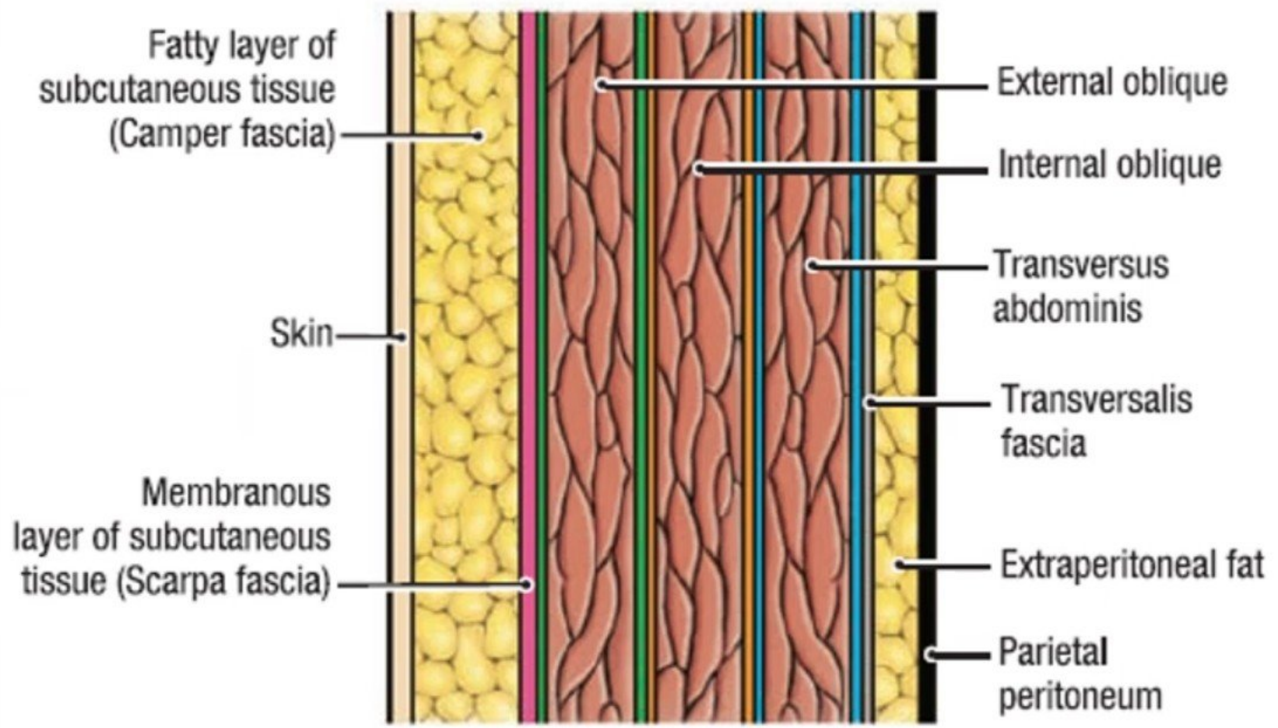
Lecture points

- Anterior abdominal wall have 8^{th} layers (remember).
- Definition of Inguinal canal.
- Inguinal canal orientation.
- Inguinal ligament.
- Inguinal canal like (a Box !!).
- Deep inguinal ring (Inlet of inguinal canal)
- Superficial inguinal ring
- (Outlet of inguinal canal)
- The anterior and posterior wall of the canal.
- The roof and floor of the inguinal canal.
- The integrity of the inguinal canal depends upon:
- The Contents of inguinal canal.
- The round ligament of the uterus in females in the inguinal canal
- The ilioinguinal nerve.
- The Main functions of Inguinal canals.
- The Defensive mechanism (shutter mechanism).

- Anatomical dissection of inguinal region in male
- Anatomical dissection of inguinal region in female.
- The Spermatic Cord.
- The layers coverings of the spermatic cord
- The contents of the cord consist .
- Descent of gonads in male .
- Descent of gonads in female.
- Applied clinical anatomy.
- Indirect (oblique hernias)
- Direct hernias.
- Pressures in the inguinal canal
- Inguinal triangle.
- Relation of the types of hernias to inguinal triangle
- Summary of clinical correlation
- testicular venogram.

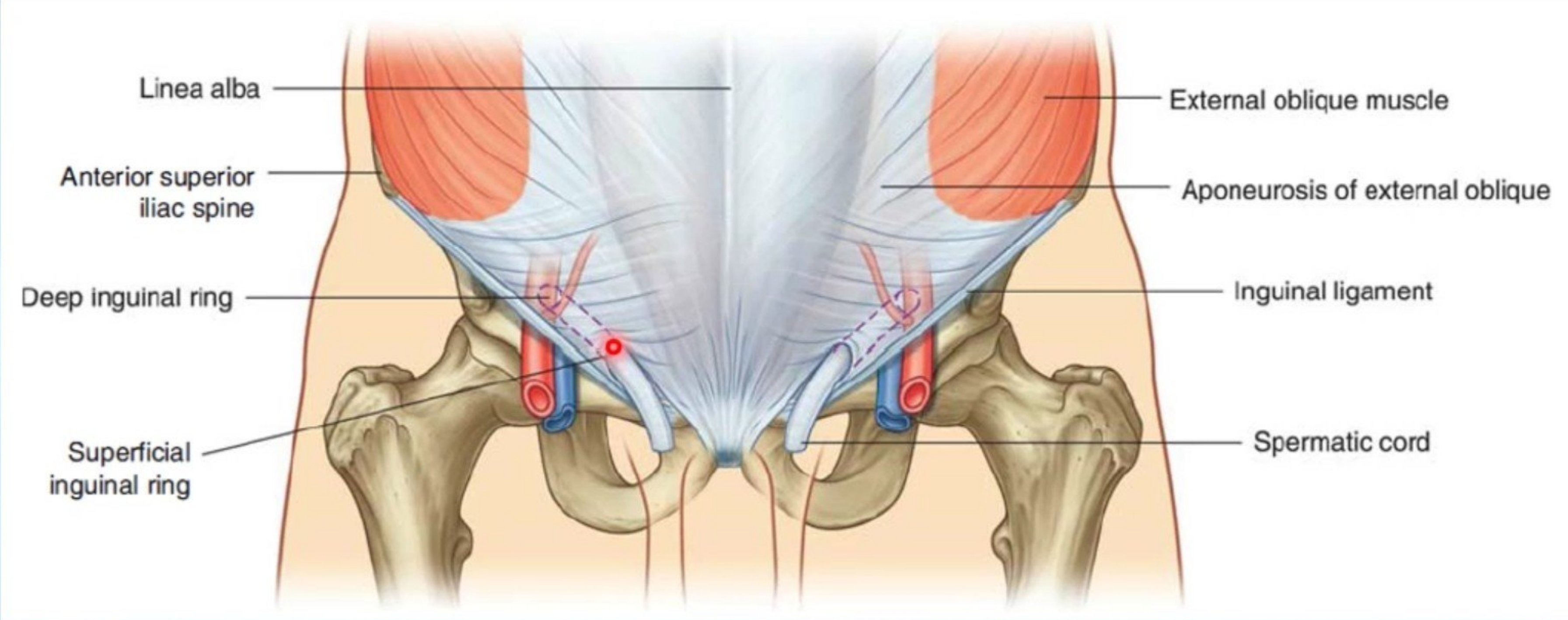
Anterior abdominal wall have 8 layers:

1. Skin.
2. Superficial fascia.
3. External oblique muscle.
4. Internal oblique muscle.
5. Transversus abdominis muscle.
6. Fascia transversalis.
7. Extra peritoneal tissue.
8. Parietal peritoneum.



C. Longitudinal Section

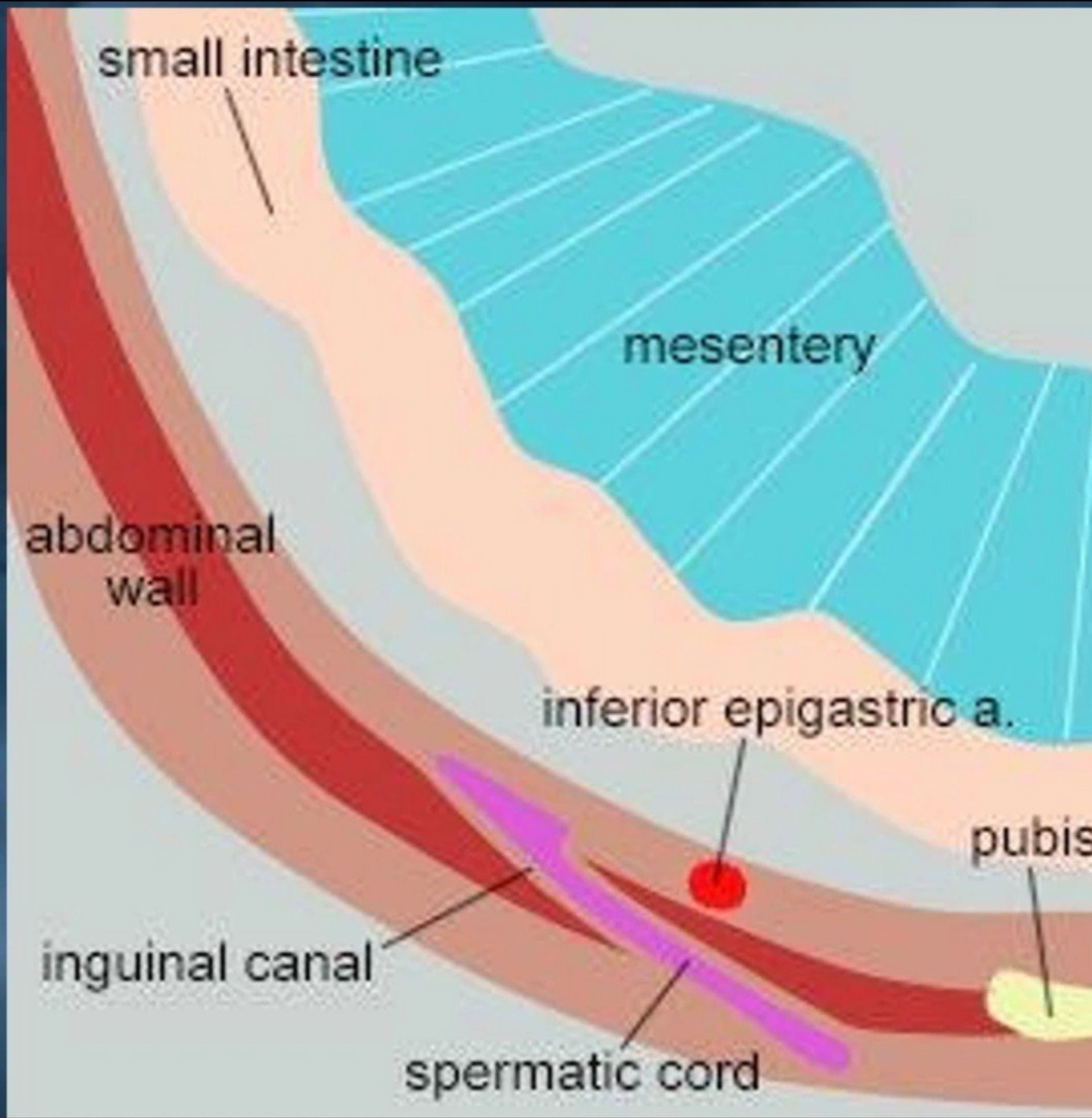
Inguinal canal



Definition :

- It is a **musculoaponeurotic tunnel** extending from **superficial inguinal ring** to the **deep inguinal ring**.
- The inguinal canal is about **4 cm long** and
- It is **directed obliquely** and inferomedially through the inferior part of the anterolateral abdominal wall.
- The canal **lies parallel** to the **medial half of the inguinal ligament**.

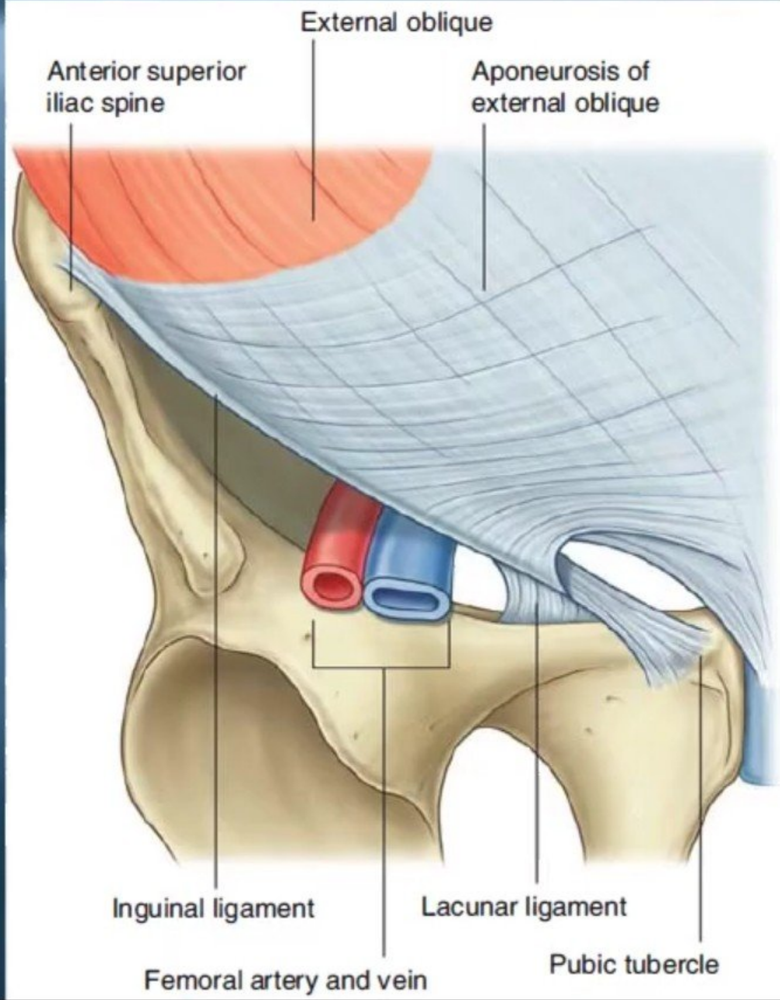
Inguinal canal orientation

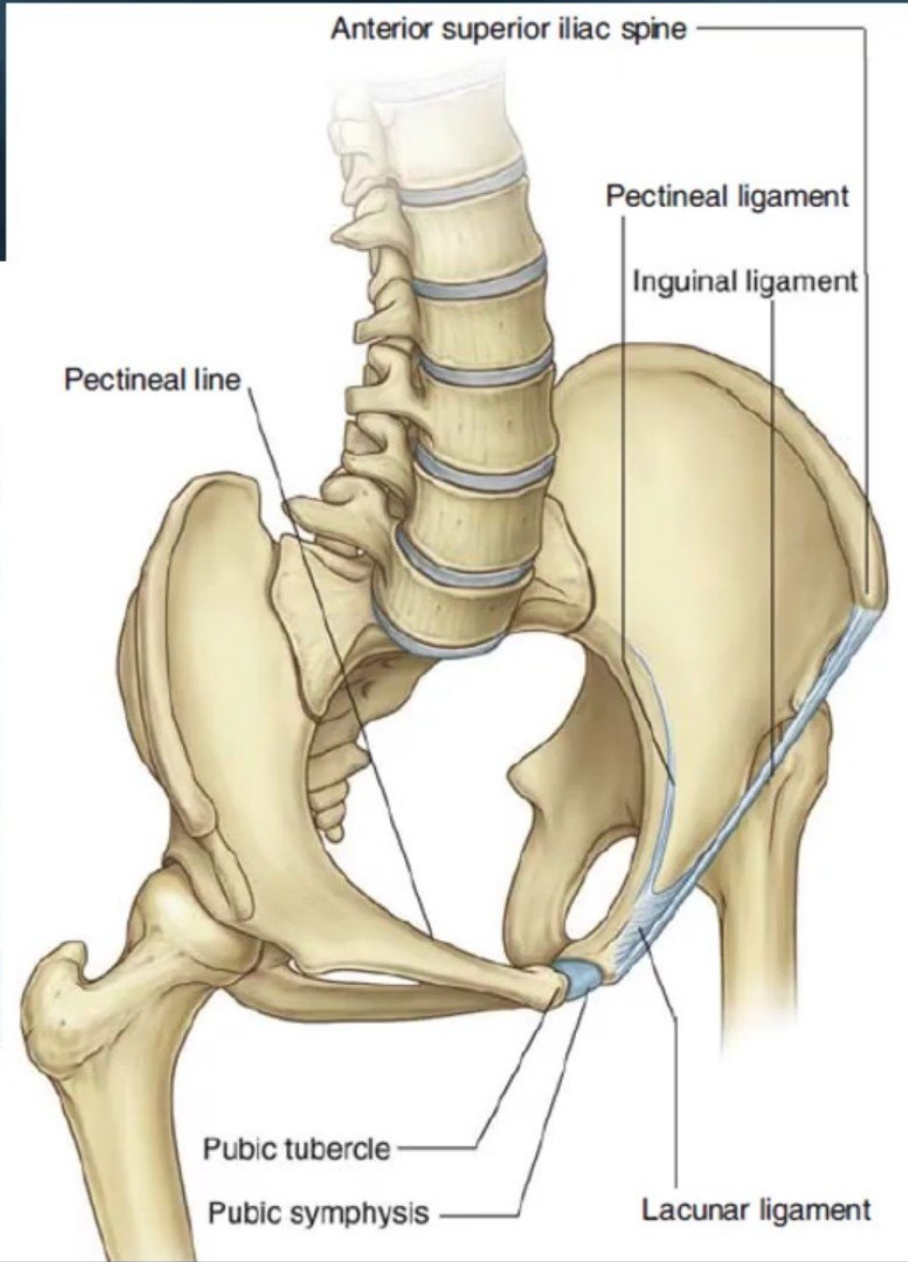


Inguinal ligament

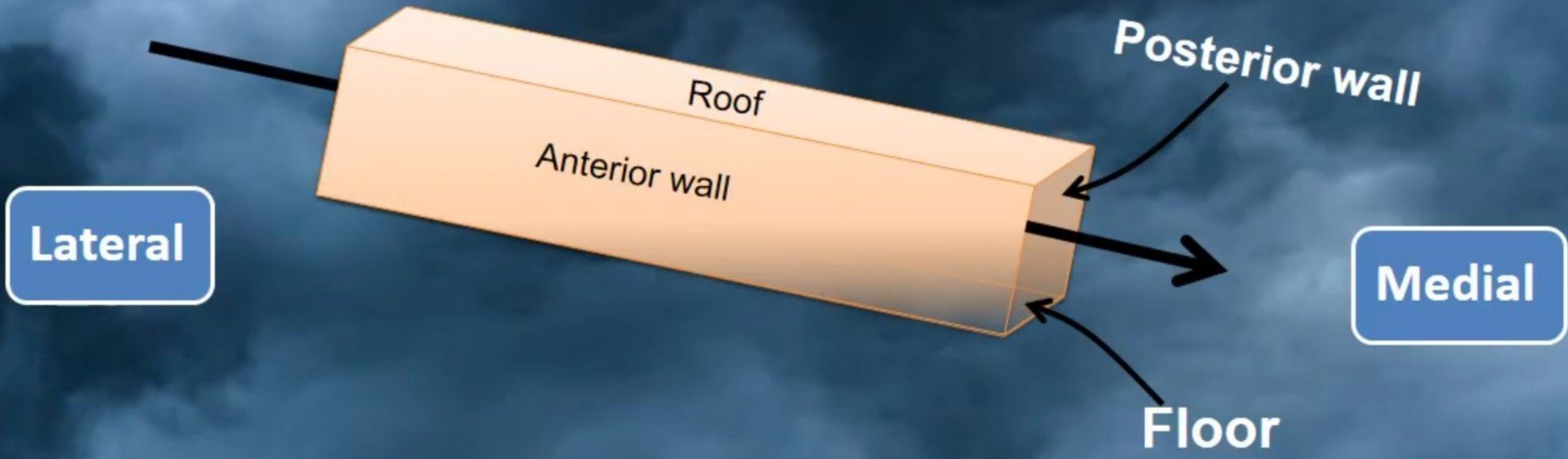
- This ligament extends from **the anterior superior iliac spine** to the **pubic tubercle**.

- It is the lower free edge of **the external oblique aponeurosis**.





Inguinal canal like (a Box !!)



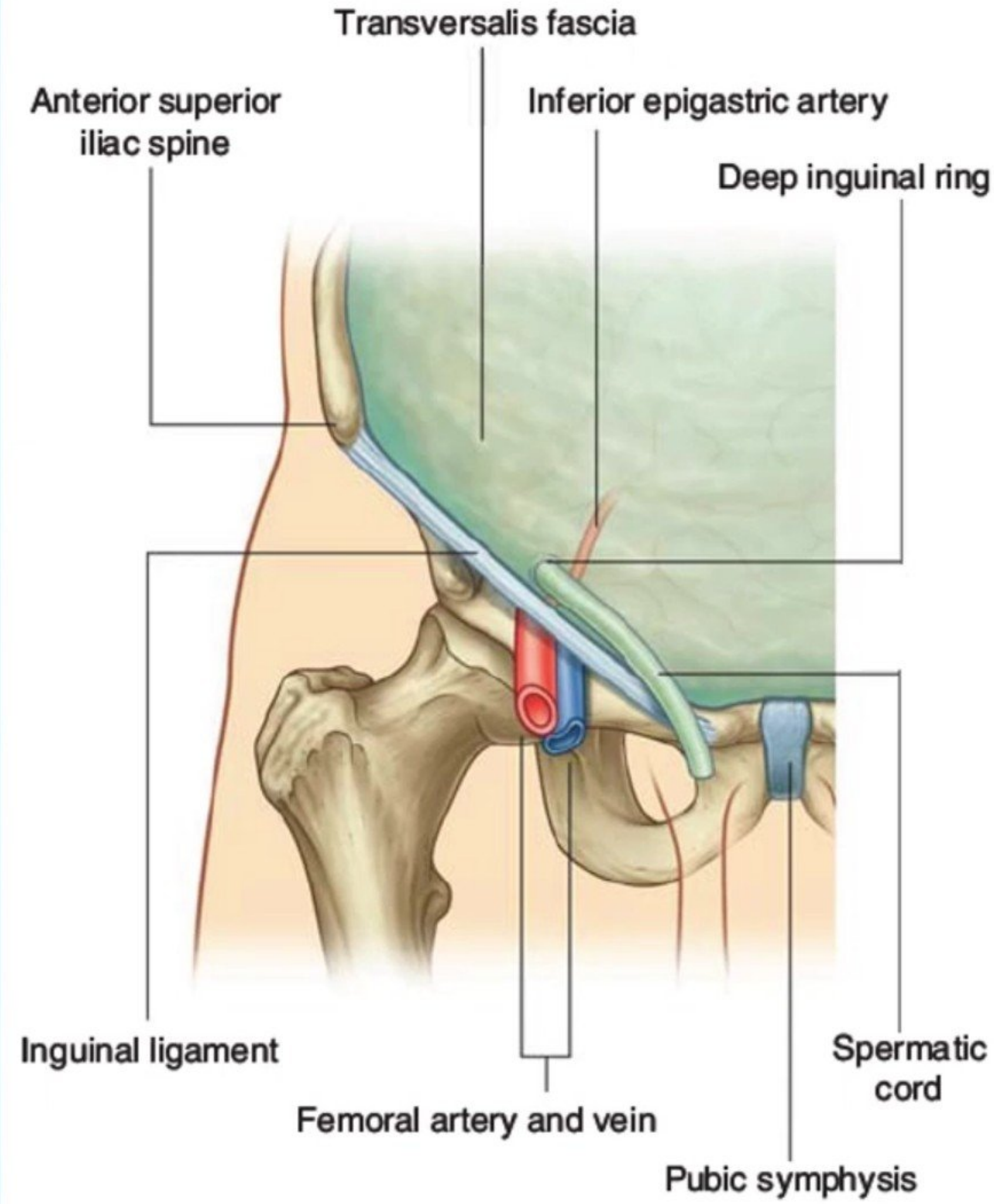
Imagine the right side inguinal canal viewed from the front as a box with anterior & posterior walls, a roof & floor. The arrow indicates that structures can run through it from lateral to medial – e.g. in males it transmits the spermatic cord, and in females, the round ligament of the uterus.

Deep inguinal ring
(Inlet of inguinal canal)

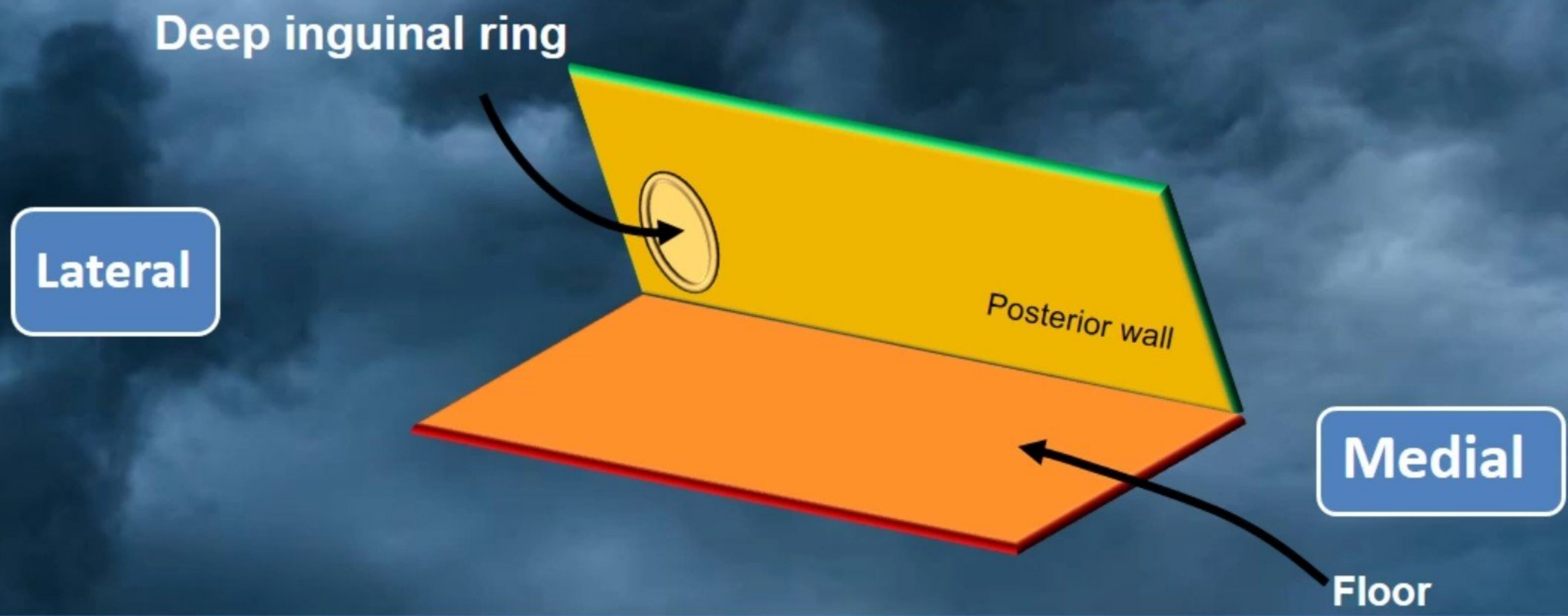
- **The inguinal canal has openings:**
  - The deep inguinal ring.
  - The superficial inguinal ring.

The **deep inguinal** ring is the entrance to the inguinal canal.

- It is an **oval gap** in the **transversalis fascia**.
- It is about **1.25 cm above the midpoint of the inguinal ligament**.
- From the margins of this opening the transversalis fascia is projected along the canal, like a sleeve, **the internal spermatic fascia**, around the structures that pass through the ring.

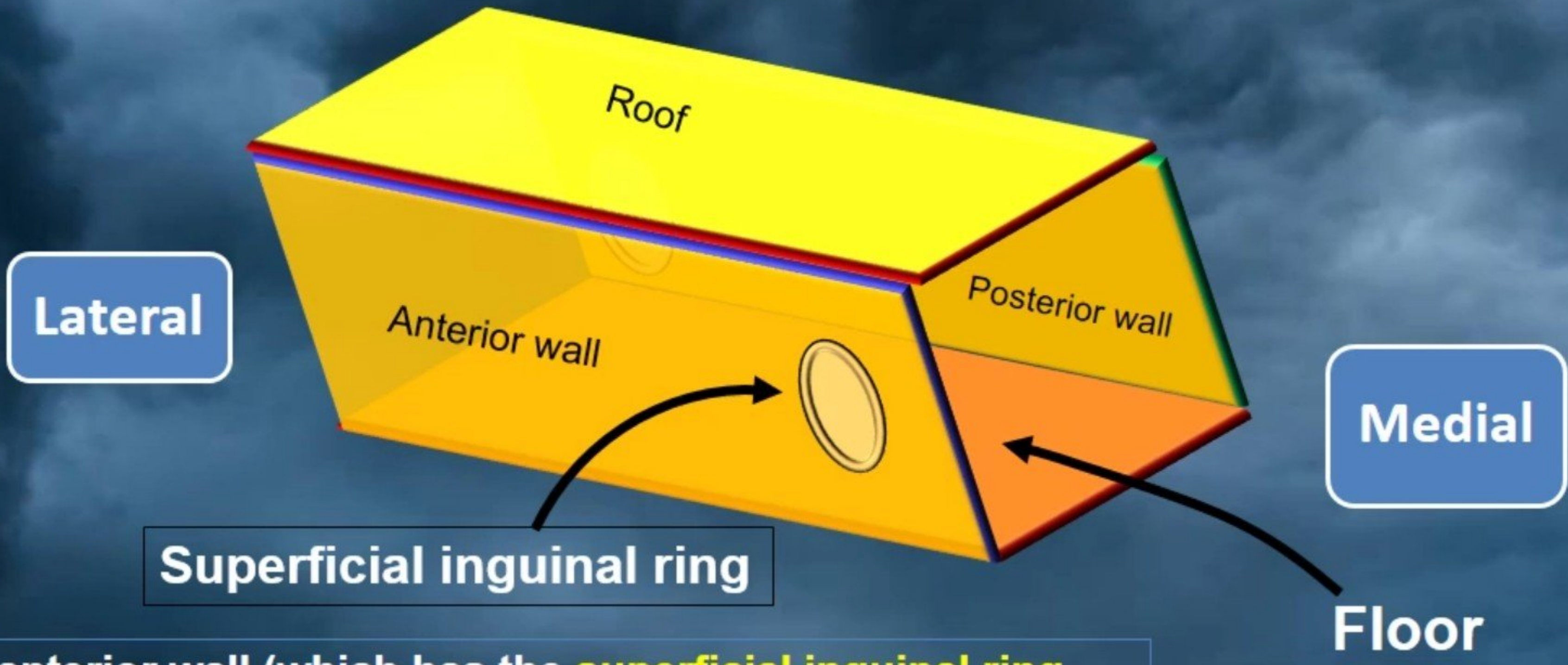


Deep inguinal ring
(Inlet of inguinal canal)



Here are the posterior wall, which has **the deep inguinal ring** situated laterally, and the floor. (Roof and anterior wall removed).

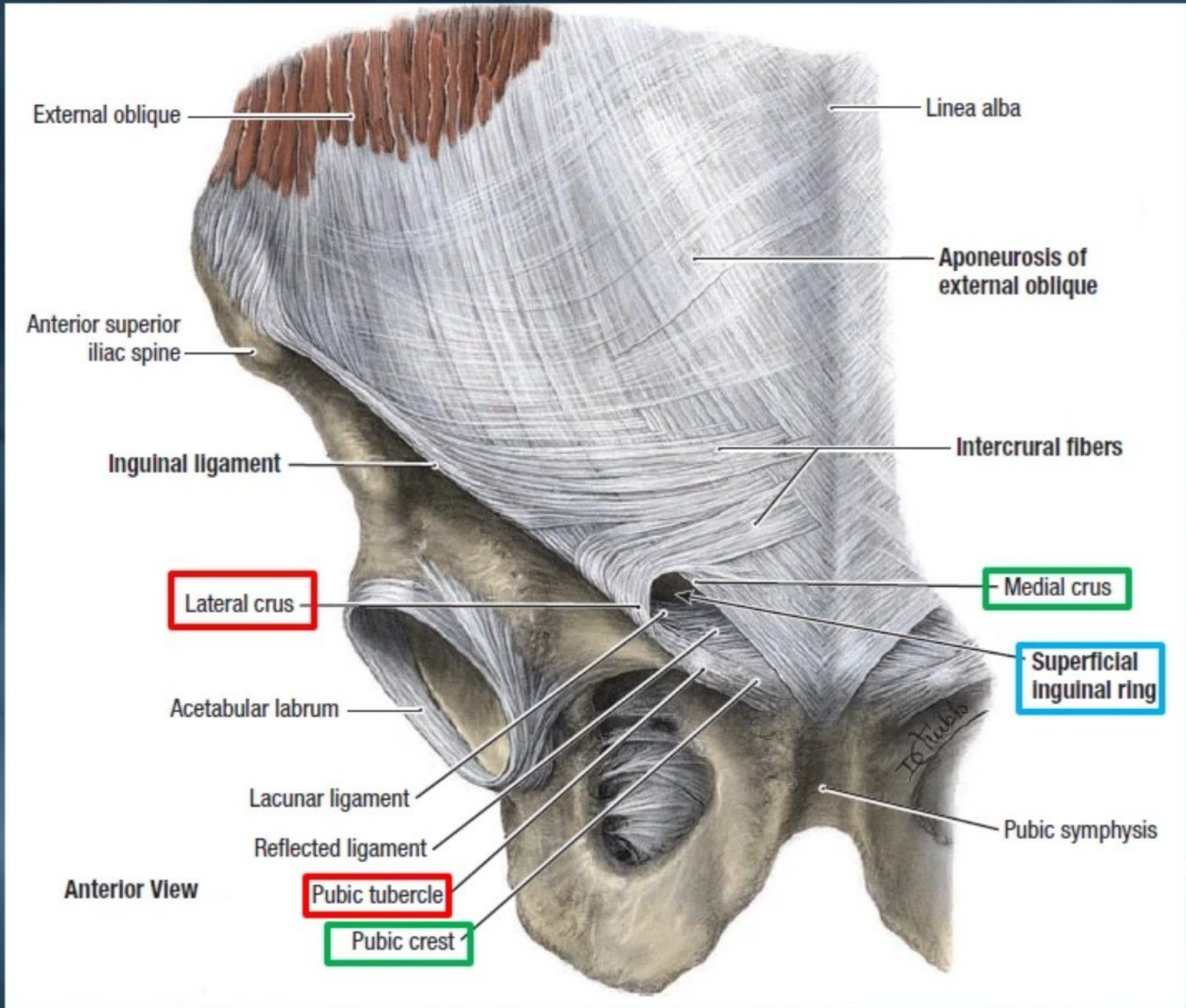
Superficial inguinal ring



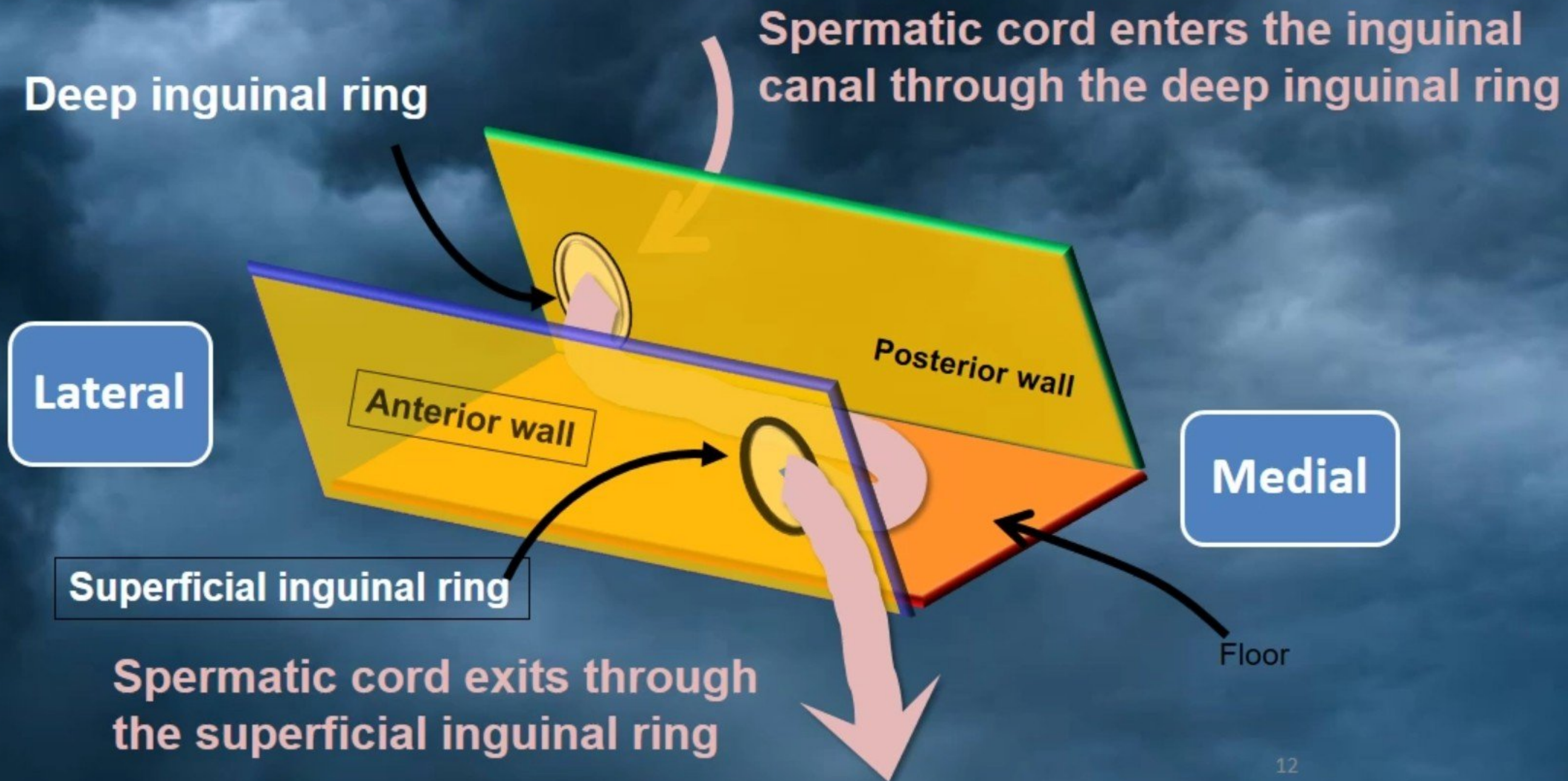
Here are the anterior wall (which has the superficial inguinal ring situated medially), and the roof.

(Outlet) superficial inguinal ring

- **The superficial**, or **external inguinal ring** is **the exit from** the inguinal canal.
- It is a **slit-like opening** in the **aponeurosis of the external oblique muscle**, superolateral to the pubic tubercle.
- The **medial** and **lateral margins** of **the superficial ring** formed by the split in the aponeurosis are caller **crura**.
  - ✓ **The lateral crus** is attached to **the pubic tubercle**.
  - ✓ **The medial crus** is attached to **the pubic crest**.



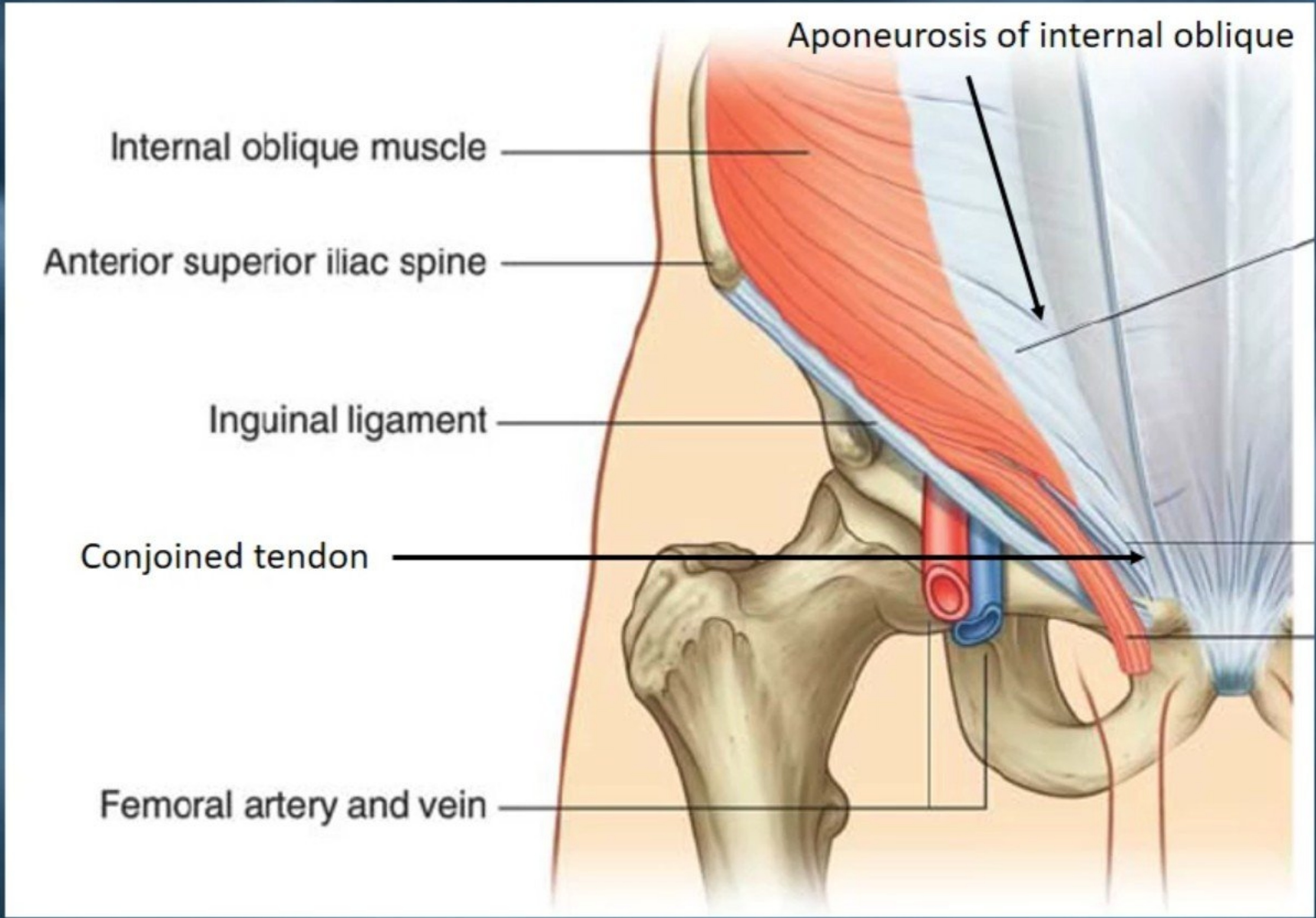
Deep and superficial inguinal ring in Inguinal canal



The anterior and posterior wall of the canal

The anterior wall of the canal:

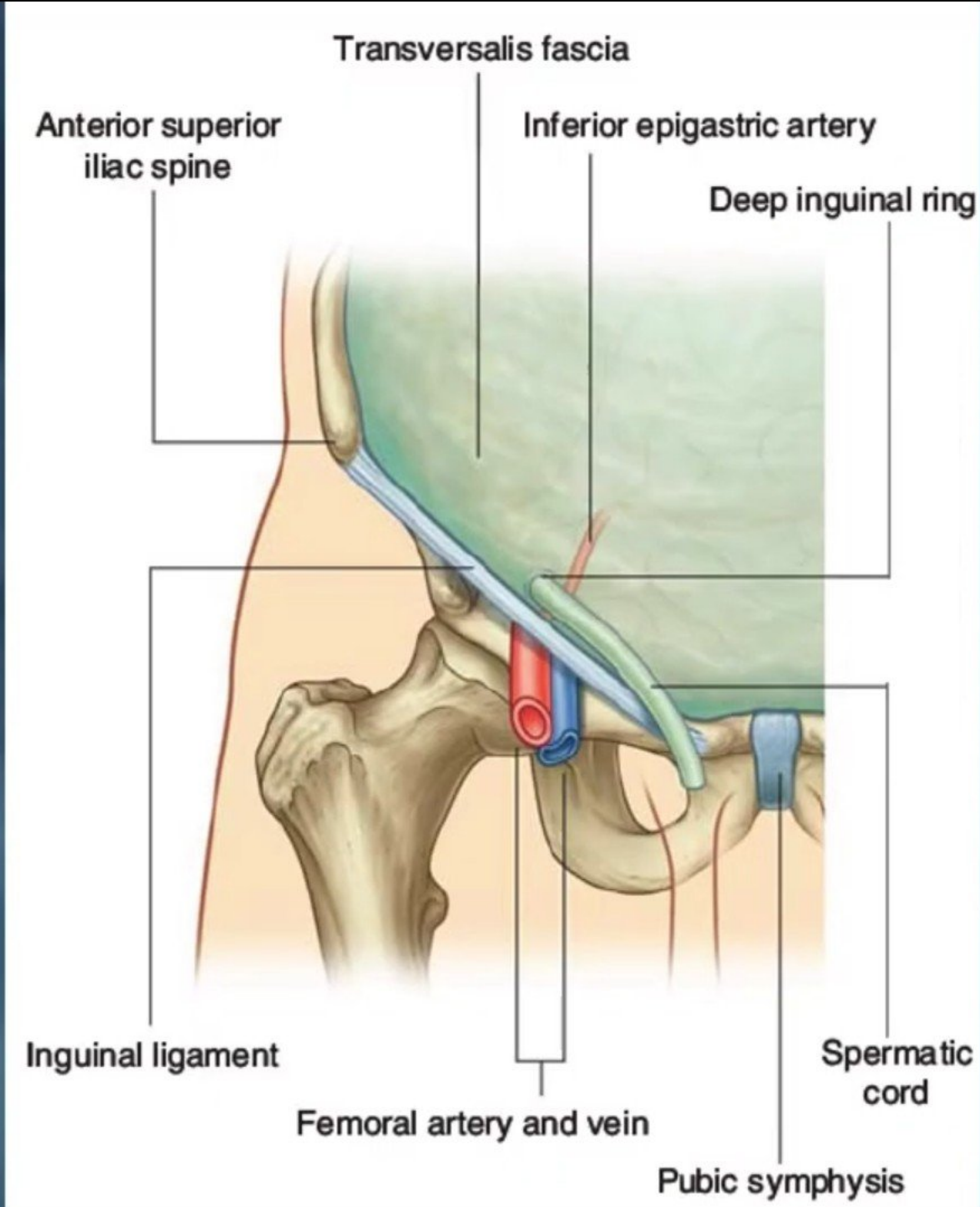
- It is formed mainly by **the aponeurosis of the external oblique**.
- **The lateral part of the anterior wall** being reinforced by fibres of the **internal oblique**.



The posterior wall of the canal

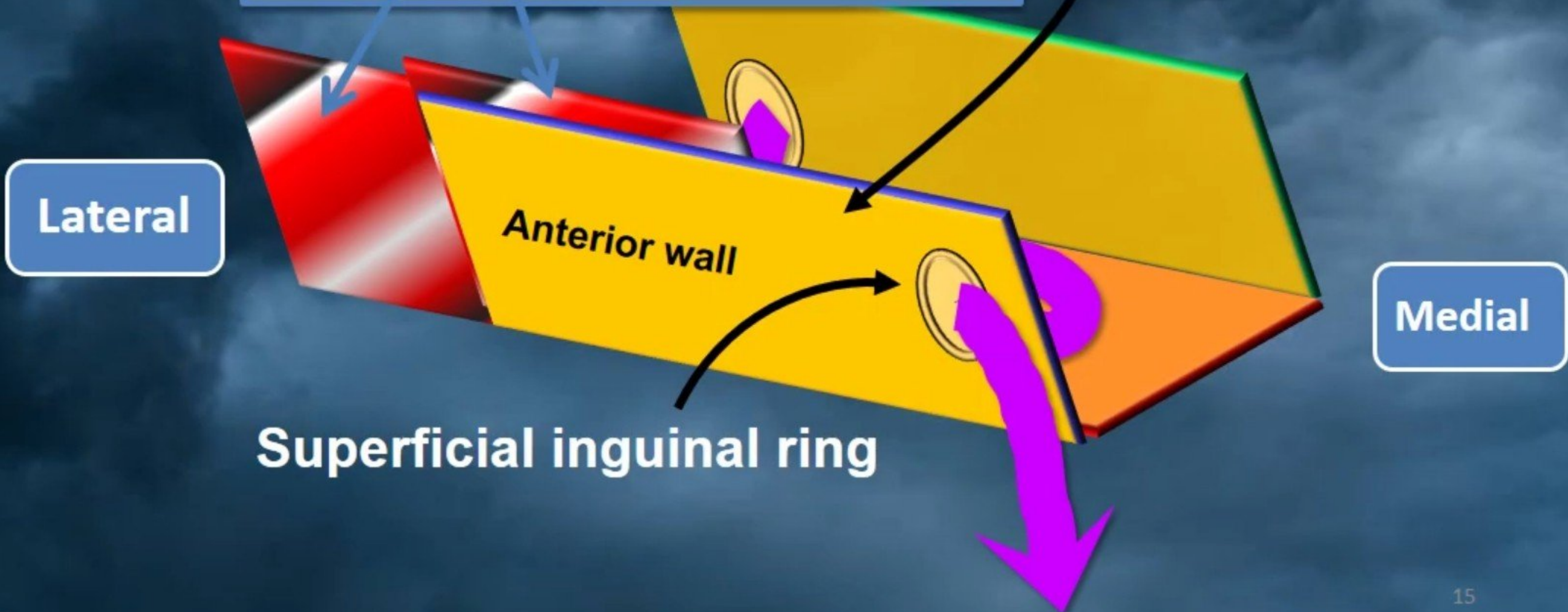
The posterior wall of the canal:

- It is formed by **transversalis fascia** with **the medial part** of the wall being reinforced by the **conjoint tendon** (the merging of the **internal oblique** and **transversus abdominal aponeurosis** into **a common tendon).**

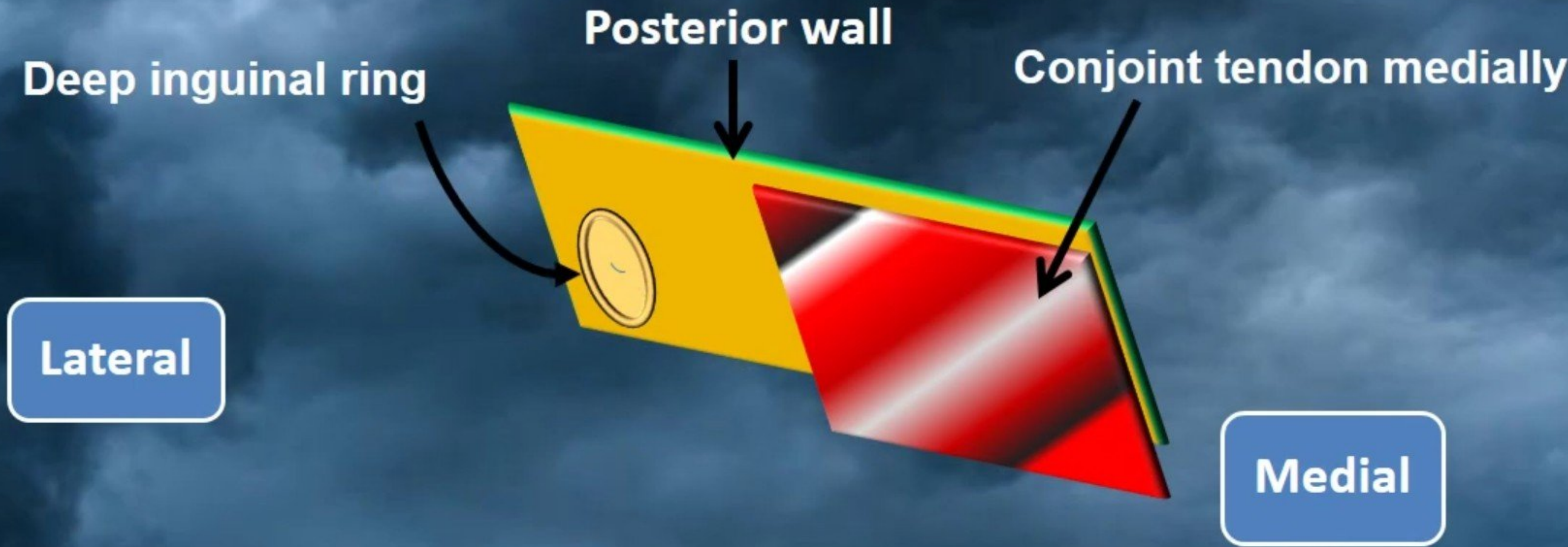


The anterior wall of the canal

The anterior wall is made up of the **external oblique** muscle throughout, and is reinforced by the **internal oblique m.** laterally.
The **transversus abdominus m.** lies even more laterally as part of the anterior abdominal wall.



The posterior wall of the canal



- ✓ The **posterior wall** is formed by **transversalis fascia (orange)** throughout and **the conjoint tendon (red) medially**.
- ✓ The **posterior wall** is particularly **weak** over the **deep inguinal ring.**

The roof and floor of the inguinal canal

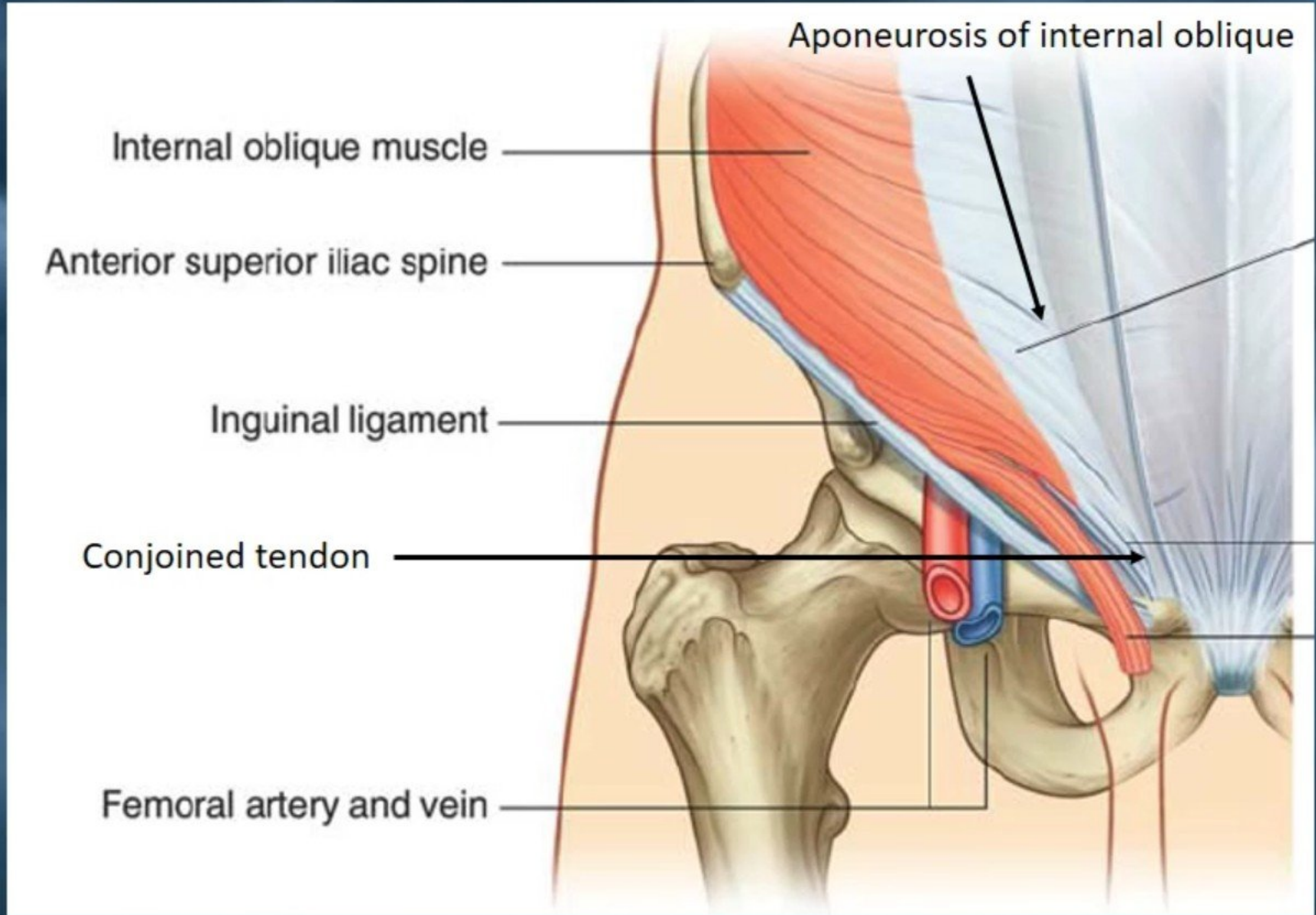
The roof of the inguinal canal is formed by:

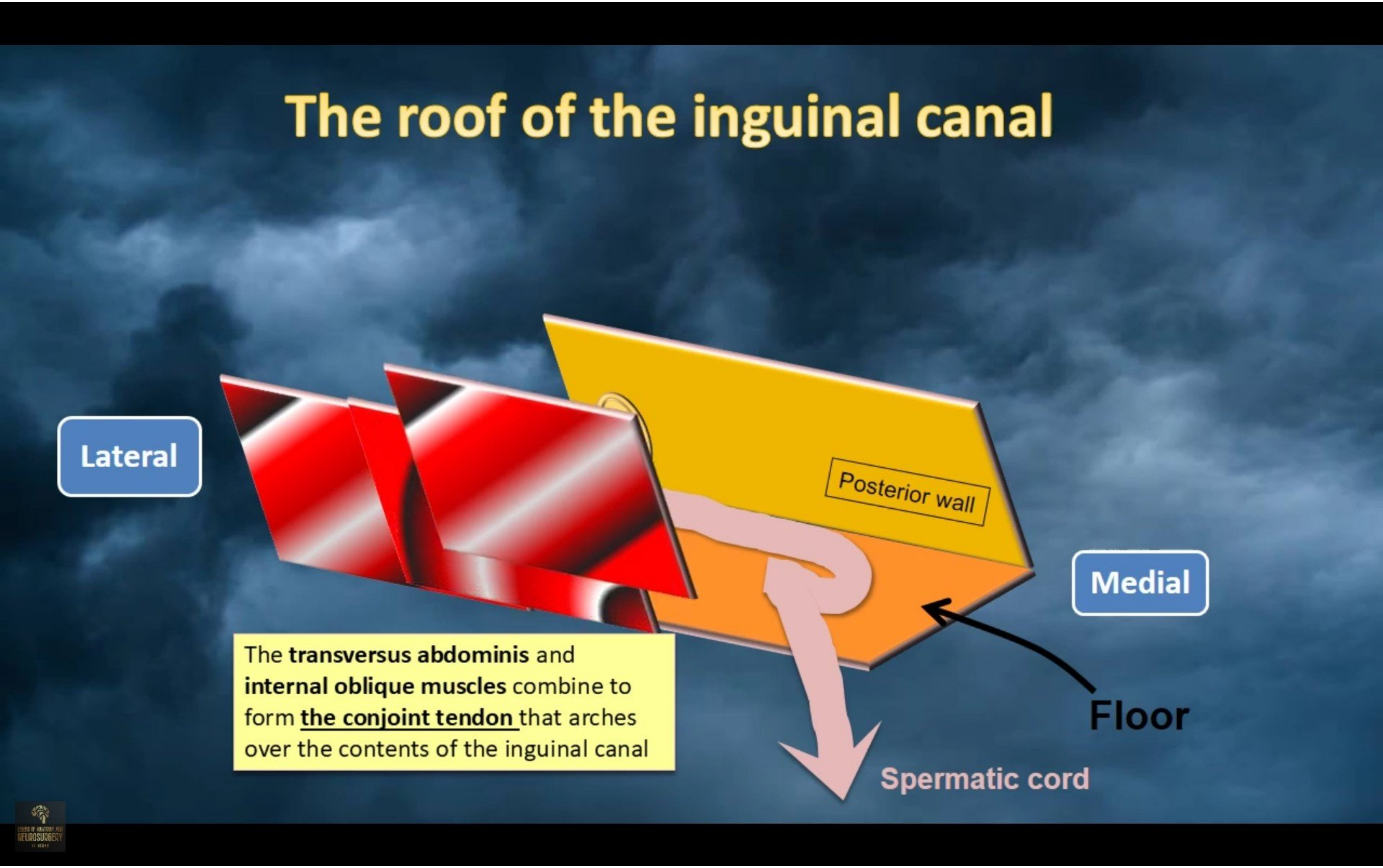
- ✓ **The arching fibers of the internal oblique muscle.**
- ✓ **The transverse abdominal muscles.**

both are forming the conjoined tendon.

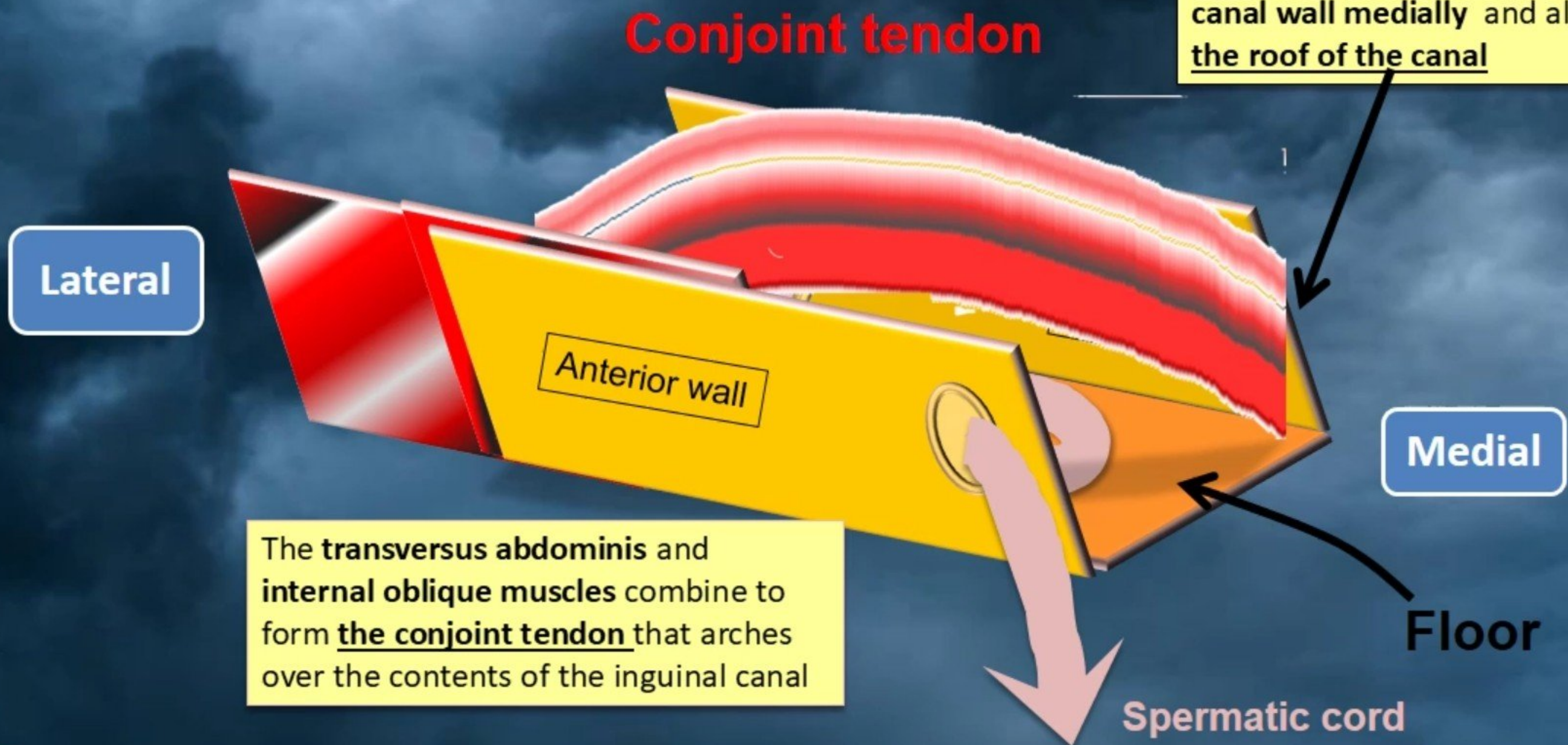
The floor is formed by:

- ✓ The superior surface of the **inguinal ligament**.
- ✓ It is reinforced in **the medial part** by **the lacunar ligament** (a reflected part or extension from the deep aspect of the inguinal ligament to the pectineal line of the superior pubic ramus).




The roof of the inguinal canal
Lateral
Posterior wall
Medial
The **transversus abdominis** and **internal oblique muscles** combine to form **the conjoint tendon** that arches over the contents of the inguinal canal
Floor
Spermatic cord

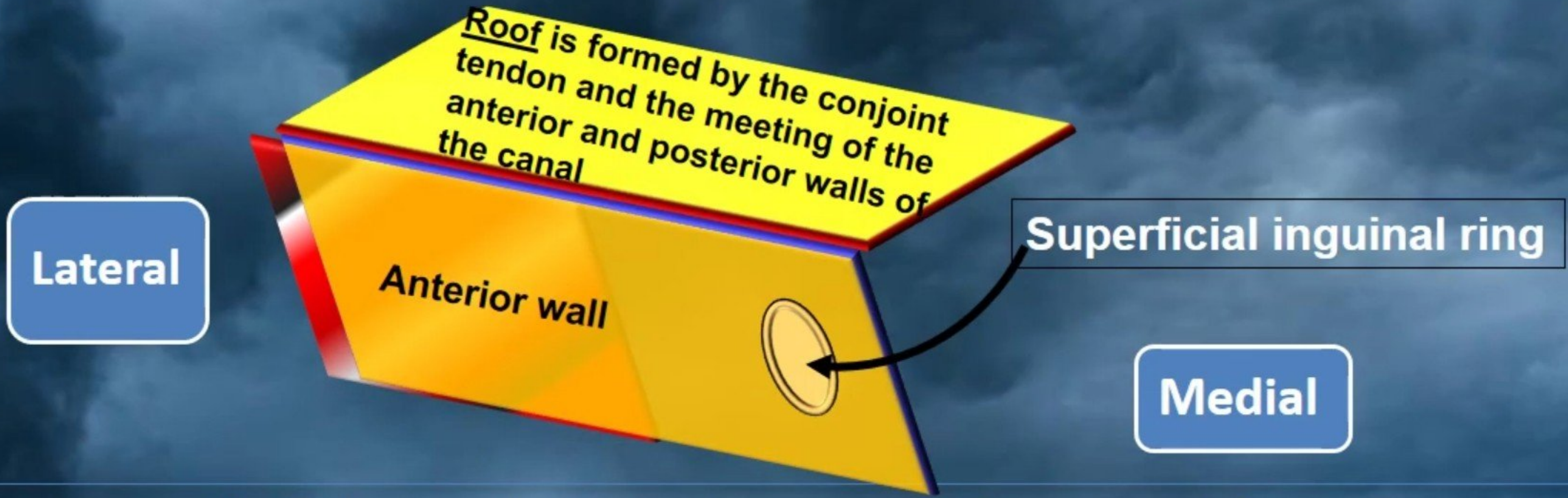
The roof of the inguinal canal



The conjoint tendon attaches to the pubic crest, **reinforces the posterior canal wall medially** and also forms **the roof of the canal**

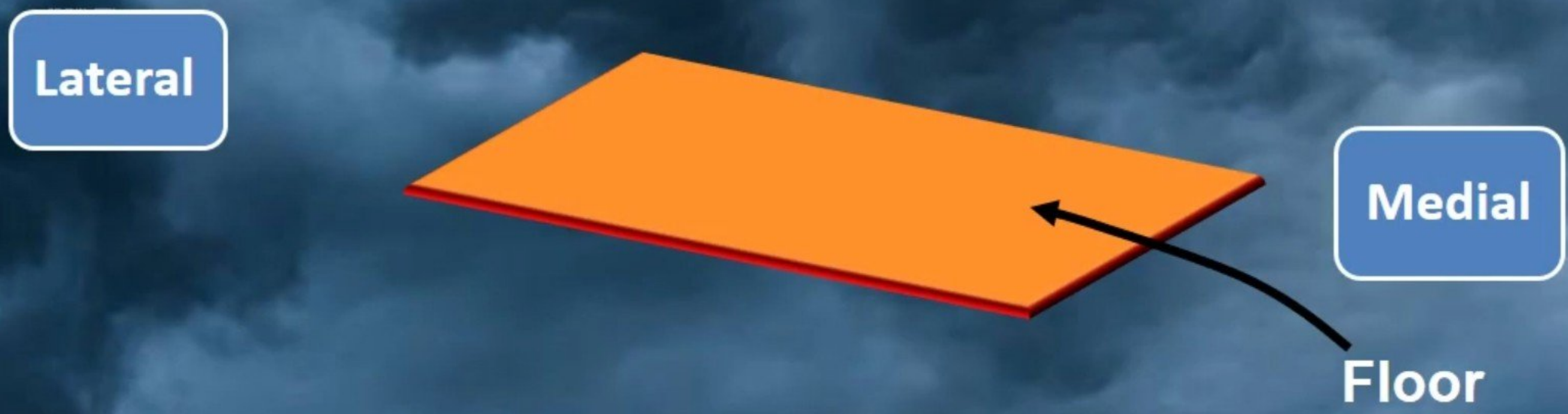
The **transversus abdominis** and **internal oblique muscles** combine to form **the conjoint tendon** that arches over the contents of the inguinal canal

The Roof and anterior wall of the inguinal canal



The anterior wall of the canal is formed by **external oblique muscle (orange)** throughout and by **internal oblique muscles** (red/black/white) laterally. This wall is weak medially because of the "hole" in the external oblique muscle (= superficial inguinal ring).

The Floor and anterior wall of the inguinal canal



The floor is formed by an incurving of the inguinal ligament, which is part of the external oblique muscle, forming a gutter. (Medially it forms the lacunar ligament which is not illustrated).

The integrity of the inguinal canal depends upon:

1) **The muscles strength of the inguinal canal depend on:**
   - ➢ **The strength of the anterior wall in the lateral part.**
   - ➢ **The strength of the posterior wall in the medial part.**
   - ✓ provided the abdominal muscles are of good tone and their aponeuroses unyielding.
2) **The deep** and **superficial inguinal rings** lie **at opposite ends** of the inguinal canal.

The integrity of the inguinal canal depends upon:

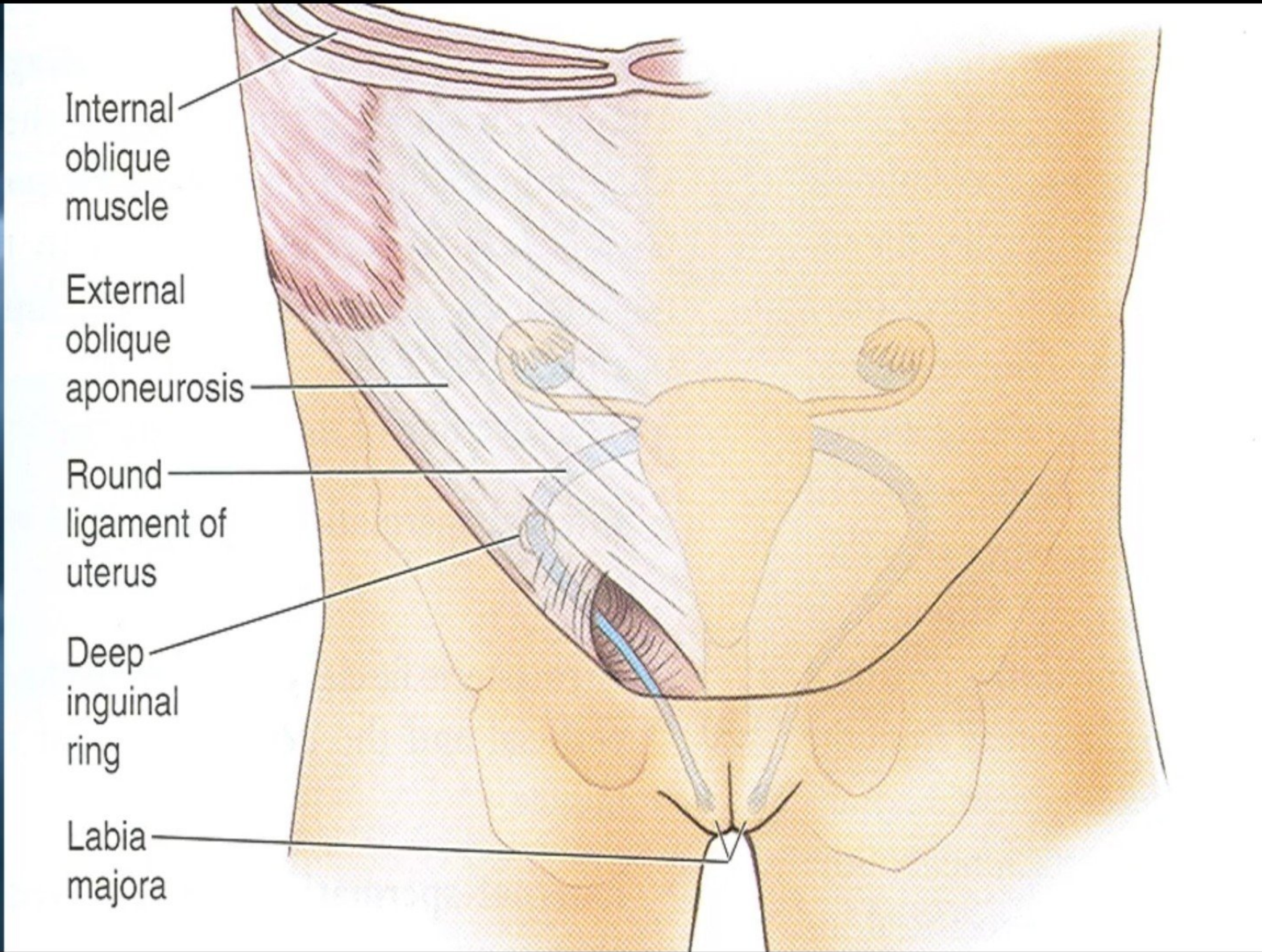
3) The **intervening part of the canal** is **pressed flat** when the aponeuroses are **under tension** and the **intra-abdominal pressure raised**.
4) The **conjoint tendon** lies **posterior to the superficial inguinal ring** and helps to reinforce this area.
5) **Laterally the transversalis fascia** in **the posterior wall** is strengthened by the presence in front of it of tendinous, and sometimes muscular, fibres derived from the transversus abdominis muscle.

The Contents of inguinal canal

The main content of the inguinal canal is:

1) The **spermatic cord** in **males**.
2) The **genital branch** of the **genitofemoral nerve**.
3) The **round ligament** of the **uterus** in **females**.
4) The **ilioinguinal nerve L1**.
5) The canal also transmits the **autonomic nerves,** the **blood and lymphatic vessels.**

The round ligament of the uterus in females in the inguinal canal



The ilioinguinal nerve

- **The ilioinguinal nerve**, although a content of the inguinal canal, **does not enter** the canal through **the deep inguinal ring**, but **by piercing the internal oblique muscle**.
- It lies **in front of the cord** and **leaves** the canal through **the superficial ring**
- It is supply skin of :
  - ✓ The **inguinal region**.
  - ✓ The **upper part of the thigh**.
  - ✓ The **anterior third of the scrotum** (or **labium majus**).
  - ✓ The **root of the penis**.

The Main functions of Inguinal canals

Main functions of inguinal canal:

- Allow **contents of the scrotum** to **communicate** with **intra-abdominal contents**.
- **Prevent mobile intra-abdominal contents** (e.g. intestine) from **entering the scrotum** and possibly becoming **damaged**, while at the same time permitting blood vessels, nerves, lymphatics, vas deferens etc. to supply the scrotal contents.

The Defensive mechanism (shutter mechanism)

- The **canal passes obliquely** through **the three anterior abdominal muscles** so **the two rings are at different positions**.
- When **intra abdominal pressure is increased** the **canal closes like a flap valve**.
- The canal is protected by **two of the anterior abdominal muscles.**
- The **superficial ring** is **protected** posteriorly **by the conjoint tendon**.
- The **deep ring** is posterior to the **muscular fibres of internal oblique**.
- **Arched fibres of internal oblique** and **transversus** act like **demisphincters** and obliterate the canal by bringing the roof in contact with the floor.
- **Cremasteric plug:** The **cremaster muscle contracts** and **pulls the testis** towards the **superficial ring closing the outlet** like a **ball valve**.

Will be continue ....